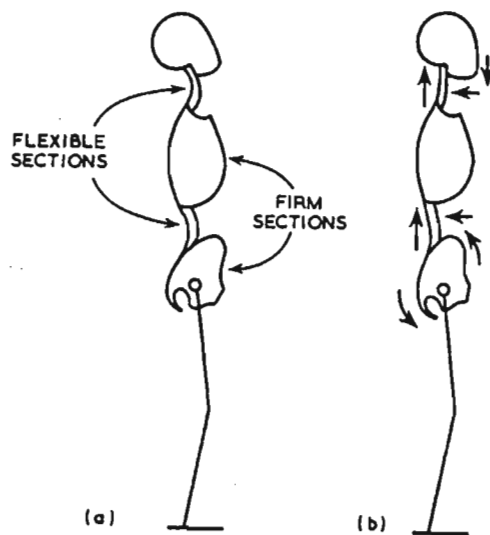


Chapter 2

Voice Production

2.1 Posture

One of the most important requirements in singing is a correct posture. Correct breathing and a good tone quality depends entirely upon the posture of the body. When choir members are allowed to sing with poor postures, incorrect vocal habits result and are deepened. Breathing can only be free and unimpeded where the adjustment of the spine and pelvis provides the correct posture. In the following diagram figure (a) illustrates the incorrect position of the lower back and the head, while figure (b) illustrates the position of the spine after correct adjustments have been made¹.



The human body tends to curve at the small of the back. This should be adjusted by pulling the buttocks in to be in line with the spine. As a result the lower abdominal wall draws in slightly. When in a standing position, the singer's feet should be planted approximately 20 centimetres apart, with the one foot slightly in advance of the other. The weight of the body should rest to a great extent on the balls of the feet.

¹Ibid., p.15.

When the singers are seated, the correct posture from the waist up will be almost identical with the standing posture. Paul Roe suggests that the singers "sit tall" with the body lifted out of the hips¹. Both feet should be kept on the floor, and the singers should lean slightly forward in order to maintain adequate breath support. Some of the body weight should be distributed to the lower limbs and the feet. The chest should be held raised without being rigid. The head should rest freely on the end of the spine, so that the muscles of the neck, the throat and the larynx are without tension. The hands should hang loosely at the sides, or be lightly clasped.

The music must be held in such a position that the director may be easily followed without the singers lowering or raising their heads, because deviation in the position of the head is likely to affect the tone quality. Paul Roe² illustrates the gravity of poor posture by using the following example: "The body is an instrument. If this personal instrument were a trumpet, what would happen to the tone quality if you took hold of the mouthpiece and bent the neck of the instrument over to a 45° angle?".

2.2 Breath Control

The importance of proper and adequate breathing cannot be over-emphasized in choral singing. The conductor should continually emphasize the correct procedure, and be alert at all times to point out incorrect breathing methods. No concept has enjoyed such universal agreement in vocal pedagogy as the importance of breath control. Three distinct methods of breathing exists, and they are:

- (a) Clavicular breathing. The shoulders, the collar bones (clavicles) and the upper part of the chest are raised. This method is generally used for speaking and reaches only superficially to the upper part of the lungs near the collar bone.
- (b) Lateral or Intercostal breathing. In this form of breathing the ribs are raised by the outer intercostal muscles, and in bulging

¹Paul Roe, *Choral Music Education*, (Englewood Cliffs: Prentice-Hall, Inc., 1970), p.75.

²Ibid., p.73.

outwards they increase the capacity of the chest in a lateral direction. Even though the parts of the lungs which lie between the armpits and waist are reached, this is still not sufficiently deep for singing.

- (c) Diaphragmatic breathing. The diaphragm contracts, and descends upon the organs contained in the abdomen, thus producing a bulging out of the abdominal wall. This method of breathing increases the capacity of the chest from top to bottom¹.

There seems to be many different views as to the correct method of breathing. Blanche Marchesi and Norris Croker share the view that diaphragmatic breathing is the only true method while David Slater is convinced that a combination of intercostal and diaphragmatic breathing is the correct principle. He reasons that if breath is taken too deeply, it causes excessive distension of the abdomen, making it impossible to obtain a free action of the ribs and proper expansion of the chest².

In singing, only full, deep breathing should be employed, which in itself is a combination of diaphragmatic and intercostal breathing. It also protects the singer's posture and poise. Deep breathing helps to overcome stage fright and the accompanying rapid heartbeat. Anxiety of this kind causes a drawing in of both shoulders and abdomen and results in nervous tension and congested breathing.

Inhalation is brought about by the action of the diaphragm - a wide, saucer-shaped muscle which divides the upper part of the torso from the lower part beneath the ribs. In a state of rest, the diaphragm is curved slightly upwards. During inhalation it flattens and spreads out somewhat, causing the ribs to expand. This permits the outside air to rush into the lungs. The lungs should expand naturally in every direction, in the same way as when a sponge is put into water and swells out on every side³. The shoulders should never be raised during

¹Adelheid Armhold, *Singing*, (Cape Town: Tafelberg-Uitgewers, 1963), p.26.

²David Slater, *Vocal Physiology and the Teaching of Singing*, (London: Lowe and Brydene Ltd., n.d.), p. 31.

³Ernest White, *Science and Singing*, (London: J.M. Dent & Sons, Ltd., 1938), p. 124.

inhalation, and should remain relaxed and slightly drawn back and down. This represents an alert, prepared position.

The singer's breath training does not begin with inhalation, but with exhalation. It is the conscious, relaxed but carefully controlled, outgoing breath stream upon which all good singing activity rests. This even flow of outgoing breath should not be constricted at the throat and should be controlled by the abdominal muscles. As previously mentioned¹, support is provided by the abdominal breathing muscles working in contrasting motion to the diaphragm and intercostal muscles which, in turn, govern inhalation.

Quiet breathing is most important. The Latin proverb, *ars est celare artem*, (the art is to hide art) holds true regards singing as much as any other skill². An expression of wonder, surprise, (imagine smelling a flower), produces an open throat. By imagining that the breath rushes noiselessly into the lungs, the whole singing mechanism assumes the relaxed and open position which is necessary for the moment of sound production.

Many famous singing tutors believe in drawing breath through the nose³. Breathing deeply through the nose, however, is too slow a process for singing. A sleeping person inhales through the nose which then acts as a filter to prevent heat, dust or insects penetrating the throat and lungs. During speaking or singing, another method of inhalation becomes necessary, and breathing through the nose alone becomes impossible. The most logic method seems to be borne out of necessity; in order to fill the lungs quickly and efficiently, and not to break the rhythmic continuity of the music, singers have to inhale rapidly through the mouth and nose simultaneously.

Breath training should become an integral part of choral rehearsals. Not only does it stimulate breathing itself, but it also serves to prepare the singer psychologically. As a result of good breathing, the

¹Part one, p. 4.

²Ibid., p. 124.

³Wilhelm Ehmann, *Choral Directing*, (Minneapolis: Augsburg Publishing House, 1968), p.22.

singer will develop his ability to phrase artistically and intelligently¹.

Simultaneous breathing is one of the essential means toward achieving a vital, corporate unity within the choir. The breath taken at the beginning of a song unifies the choir into an animated, close unit and so helps to ensure precise and correct entries. The director breathes with the choir and accompanies this procedure with an appropriate gesture which leads directly into the first beat of the music. Very often, choirs suffer from "overpressure" of breath. The choir should be requested to exhale on a deep sigh and immediately proceed to use up the residual air by speaking on the syllable "dah, dah, dah". The singers will definitely be able to produce a long string of "dahs".

In choral singing, three different breathing techniques may be used: deep, snatch, and staggered breathing.

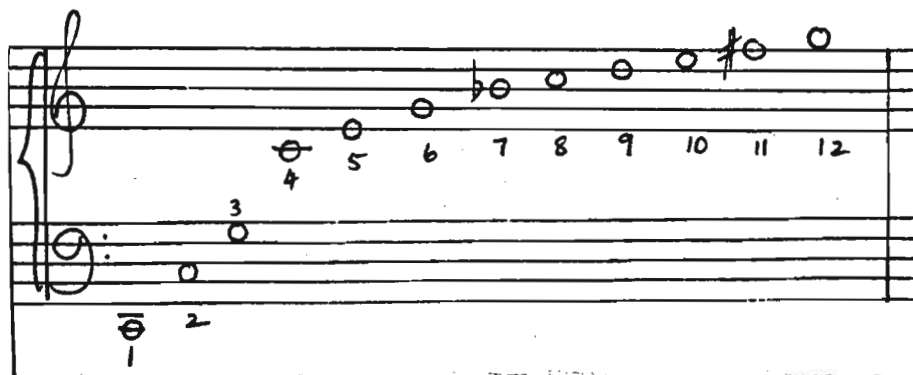
- (a) Deep breaths should be taken at the end of well-defined phrases and after half-cadences and whole cadences. If the conductor can induce the singers to breathe deeply after phrase endings, the musical structure will be made clear and the music itself will be full of life and movement.
- (b) At times the singers may have to snatch a breath which is possible through a quick, small movement of the stomach walls just beneath the solar plexus. The posture of the singer remains unchanged.
- (c) Staggered breathing must be applied whenever a phrase is so long that the individual is unable to sing it in one breath. The choir director should decide on the breathing places for the different sections. In their turn, the singers help each other by breathing in such a way that the whole phrase can be sustained without interruption and without any audible inhalation.

¹Part 2, pp. 57-58 contains various breathing exercises.

2.3 The Vocal Resonance System

The sound of the human voice is very complex and rich in harmonics, for the mouth and various cavities opening into the mouth such as the pharynx, the naso-pharynx and the chest, act as resonators.

A single vowel sound does not exist, because any vibrating body such as the vocal cords, sets up subsidiary vibrations as well. While vibrating along its entire length, the fundamental tone is produced. Almost immediately, its halves set up subsidiary vibrations and, being half the length, vibrate twice as fast, sounding the octave. The thirds vibrate three times as fast, sounding the twelfth. Eventually, we hear something like this:



This relationship of sound is called the harmonic series: the first harmonic is called the fundamental. The remaining notes or harmonics are called upper partials or overtones¹.

Nasal resonance is an added embellishment which gives a richer sound to the tone resonating in the mouth cavity. Nasal resonance should not be confused with nasal tone, which is caused by a lowered soft palate, thereby shutting the mouth off from the throat. As a result, the tone passes directly through the nose, resulting in an offensive nasal quality. Humming exercises should be practised to develop correct nasal resonance².

¹Charles Cleall, *Voice Production in Choral Technique*, (London: Novello and Company Ltd., 1970), p. 53.

²Part 2, pp. 65-67 contains appropriate exercises.

Tone production is much improved by the addition of head resonance. The effect of head resonance is a brilliance resulting from the strengthening of high harmonics. The singers' entire range should contain head resonance as it's brilliance is an important factor for singing in tune: notes without such brilliance tend to sound flat.

The mouth is a large, spacious cavity, the roof being formed by the high palate. The position and shape of the lips and mouth have the greatest influence upon the timbre of the voice.

The chest is a superb and powerful resonating box, adding great volume to the low tones of the voice. Chest vibration decreases naturally as the voice ascends the scale.

The vocal cords merely produce the sound. The mouth transforms it into tone, which, with proper cavity formation and breath control, resonates in the mouth, chest and cavities of the head. If there happens to be a cavity in the mouth, nose or chest which is "tuned" in to one of the partials produced by the larynx, that particular partial or overtone is reinforced while the other partials, in finding no sympathetic resonator remain relatively insignificant. No tone should be pushed or forced to some particular point. Properly produced tone seems to place itself, provided there is no interference with the resonance mechanism. Therefore, singers should aim for the following:

- (a) Slightly raised cheeks, forming a gently smile so as to open the inner nasal cavities.
- (b) Low-lying relaxed tongue and lower jaw.
- (c) The teeth apart, so that the cavity of the mouth remains sufficiently large.
- (d) A high relaxed soft palate, spreading out towards the sides.
- (e) A relaxed and open throat.
- (f) A foundation of good breath control.

Chapter 3

Control of the Laryngeal Mechanism

3.1 Pitch

The fundamental pitch of a voiced sound is determined by the tension of the vocal cords and the pressure of the breath against the vocal cords.

Vocal cords are capable of changing rapidly in thickness, length, and tightness. Within each vocal cord are two sets of muscles which co-operate to produce the proper amount of tension for each desired pitch¹. However, it is the vocal tract situated above the vocal cords which augments the tone and determines its timbre and quality.

Many singing tutors agree that the mental concept which precedes the sounding of a tone is of the utmost importance. By relaxing mentally, the jaw and tongue will drop automatically and the throat will be open. The mind should then conceive of a clear, vibrant tone at the correct pitch. The attack should be firm and precise, but without force or too much effort².

3.2 Volume

Any inexperienced singer with a potentially good voice is by no means ready for choral singing which demands loud singing. The development of volume and its control is so intimately associated with the control of tone quality that it is practically impossible to consider one without the other.

It should be pointed out that loud singing is not necessarily beautiful. In fact, many young singers have beautiful voices but lack the

¹Joseph Klein, *Singing Technique*, (New Jersey: D. van Nostrand Company Inc., 1967; reprint ed., 1968), p.36.

²Ivor Warren, *The Examination Candidate's Handbook on Singing*, (London: A. Hammond & Co., n.d.), p.15.

necessary volume¹. Untrained voices are often unable to sustain good tone quality with increasing vocal intensity. As soon as the singers acquire a good voice quality, the tutor should aim at building up the singing muscles. Loud, forced singing is an abomination, often resulting in disastrous vocal problems, such as a tremolo². A well-trained, experienced conductor should be able to guide the singers successfully in developing volume without loss of quality.

3.3 Dynamics

The voice is unique in its capacity of diminishing the sound of a single note from loud to soft and vice versa. Most beginners support the piano and pianissimo notes by the muscles of the throat instead of the diaphragm. There should be no pressure on the throat and neck muscles at all. A soft tone should be an exact replica of a loud one, involving no change of mechanism. Resonance is the carrying power of the tone, and the softest pianissimo must be just as resonant as a forte tone. Soft tones demand an even greater degree of intensity of the column of breath than loud tones. Demonstrate this by requesting the students to blow on 'ss' or 'ff', first forte, then piano. In forte, one will use dynamic force and consequently more breath, while in blowing piano, less breath will be used but the controlling action of the diaphragm and rib muscles will be strongly felt³. The carrying power of the voice depends largely upon the following elements:

- (a) Control of the breath
- (b) Forward placing of the sound
- (c) Full and correct use of the resonators
- (d) Distinct articulation⁴

¹Viktor Fuchs, *The Art of Singing and Voice Technique*, (London: Calder and Boyars, Ltd., 1963; reprint ed., 1967), p.106.

²Refer to Part 2, p. 97 for exercises.

³Adelheid Armhold, *Singing*, (Cape Town: Tafelberg-Uitgewers, 1963), p.52.

⁴David Slater, *Vocal Physiology and the Teaching of Singing*, (London: Lowe and Brydene Ltd., n.d.), p.55.

In order to control a crescendo or diminuendo it is very important that the singer's ear is sufficiently trained in keeping exact vowel form and tone colour fixed in his memory. If not, a crescendo or a forte passage may be mistaken for an increase in dark tone colour and a diminuendo or piano singing with an exaggerated thinning out of the voice¹. In diminishing the tone, the opening of the throat remains the same. Only the quantity of breath being expelled is diminished by means of the diaphragm muscle. *Messa di Voce* is a technical term applied to the art of swelling or diminishing the tone by imperceptible gradation from the softest attainable piano to full volume, and vice versa². Singers should not start working on *Messa di Voce* unless they are guided by the two guardian angels of the voice - resonance and breath control³.

3.4 Colour

Light and shade constitute the principal element of expression; and the artist who does not know how to put a variety of colour into his singing will not be able to communicate with his audience. Tone colour and resonance are inseparable. Without good resonance there can be no colour.

Facial expression has an important bearing upon the colour or quality of the voice. For example, it is impossible to produce bright, happy tones with a sombre facial expression, and equally impossible to portray sad emotions with a meaningless smile.

A balanced tone has the correct blend of the fundamental tone and its overtones. A very wide smile may destroy the delicate balance by bringing in too much of the upper partials, while the fundamental remains subdued. A dark tone is the result of the emphasis of the

¹Brent Monahan, *The Art of Singing*, (London: The Scarecrow Press, Inc., 1978), p. 165.

²Refer to Part 2, pp. 78-79 for appropriate exercises.

³Viktor Fuchs, *The Art of Singing and Voice Technique*, (London: Calder and Boyars, Ltd., 1963; reprint ed., 1967), p. 114.

fundamental pitch. It lacks ring and has a tendency to stay locked in the throat¹

Without dynamic tone colour, singing would be dull and uninteresting. Not only should the words be sung clearly, but each word should also have that required quality of tone to make it expressive. The conductor and his singers should clearly understand the song as a whole, and endeavour to present the listener with a complete picture. By using the appropriate tone colour, a magical atmosphere - without which art is impossible- is created during the performance of a song.

¹Warren Angell, *Look up, you Singers!*, (Dallas: Crescendo Publications, Inc., 1978), p.73.

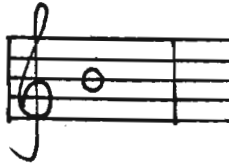
Chapter 4

Voice Ranges and Registers

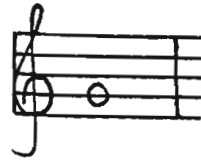
4.1 Voice Ranges

Adult singers fall into the following main categories, according to the average centre of their voices¹:

Soprano:



Mezzo-Soprano:



Contralto:

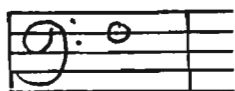


Male Alto:

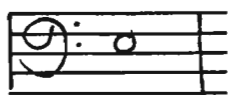


¹Charles Cleall, *Voice Production in Choral Technique*, (London: Novello and Company Ltd., 1970), p. 101.

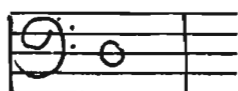
Tenor:



Baritone:

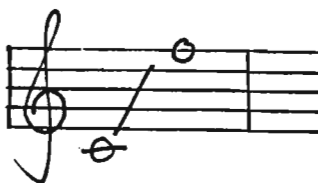


Bass:



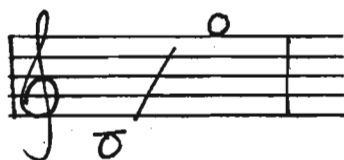
Most amateur singers range about a fifth below and above the average centre of their voices, whereas professionals will range from an octave to as much as a twelfth below and above.

The average compass of young children's voices, about 10 years of age, is approximately¹:

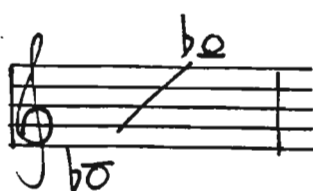


¹Reginald Jacques, *Voice Training and Conducting in Schools*, (London: Oxford University Press), p.40.

As their voices mature, with proper training, the voice range may gradually expand until the sopranos will sing quite comfortably within a range of:



Boys with good voices, given careful training, should develop a compass of:



Their voices are extremely flexible, and possess a much more extensive head register than girls.

The only way to cultivate beautiful tone in children's voices is to demand soft singing at all times. The obvious advantages are:

- (a) The voice will not be forced during soft singing, and there will be no risk of strain or fatigue.
- (b) A good tone will immediately be produced, especially in the head register.
- (c) Intonation will more often be accurate.
- (d) The registers will blend easily and naturally.

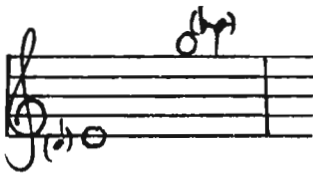
The part of the voice lying between:



should be trained first, as these are the notes on which children naturally use correct voice-production.

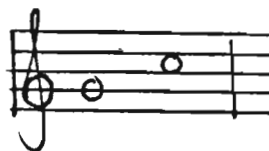
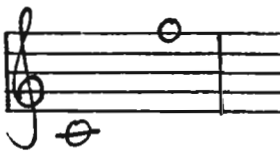
The following descriptions are applicable to the average high school choir. For each voice group, the compass is shown on the left and the tessitura on the right.

First soprano

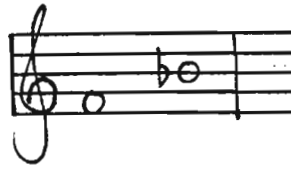
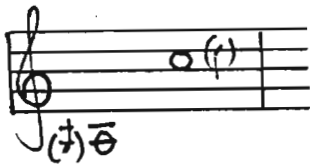


Typically, the first soprano produces a light, flutelike, lyrical sound.

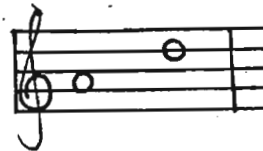
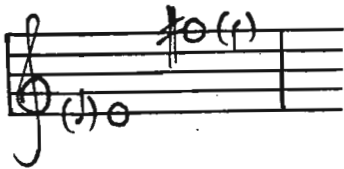
Second or mezzo-soprano



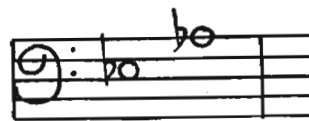
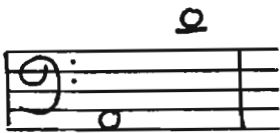
The mezzo-soprano has a fuller, more dramatic quality than the first soprano.

Alto or contralto

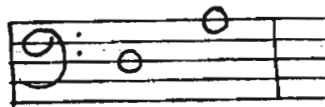
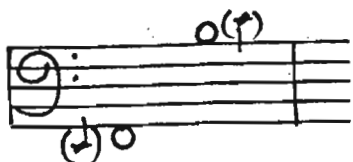
The contralto sounds full, rich and sonorous, especially in the lower range. A true alto voice is rare at the high school age, and may initially be breathy in quality.

Tenor

Characteristically, the tenor has a pure, delicate and sweet tone. While the voice sounds light in its lower range, it sounds brilliant on the higher notes where it sings comfortably. Tenors rarely mature at high school age.

Baritone

The baritone's voice is broad and rich in quality. This quality can be sustained comfortably into the tenor range.

Bass

The bass has a heavy, dark deep quality, especially in the middle and lower ranges of the voice.

4.2 Voice Registers

The distinctive character assumed by the voice, according to the particular action of the vocal organs and specific cavity employed as its "resonance chamber" constitutes what is commonly called "register"¹.

Until Garcia² invented the laryngoscope³ in 1865, all great singing teachers claimed that there were only two registers, the chest and the head. At the time of his invention, even Garcia himself believed there to be only two registers. Only at a later stage of his teaching career did he break away from the traditional view, and assert that there were three registers, chest, middle and head.

As to the idea of "one register" for the whole voice, there is no better explanation than the old Italian saying, "One register is not a starting-point, but a goal". To assert that there are no registers, is unfortunate because the blending of the registers or the smoothing out of the voice where one register passes over into another is one of the greatest problems which the teacher of voice-production has to solve. The subject of register blending is complicated because many

¹Many contradictory views are held by experts regarding this issue. One good reason for not recognising the registers, is that when they are discussed there inevitably follow references to "breaks" in the voice. This may influence the singer towards becoming self-conscious and nervous about registers and may even result in transforming a slight break in the voice to something very obtrusive.

²Manuel Garcia (1805-1906), invented the laryngoscope and was also a fine singing teacher, numbering Jenny Lind among his pupils. He was professor of singing at the Royal Academy of Music in London from 1848-1895.

³A mirror apparatus for examining the larynx.

people suppose, for example, that the vocal registers are synonymous with the different kinds of voices, and speak of the alto, soprano, bass or tenor register as if register stood for quality, which it does not.

Dr. Paul J. Moses¹, a most competent medical and vocal specialist of the Stanford University School of Medicine, writes of registers:

"The functioning of registers may be seen with the stroboscope, during mirror laryngoscopy. In singing from the highest possible tone down to the lowest, the untrained singer passes a sequence of tones of a certain character. He reaches a 'node', or switching point, from which he continues with a sequence of tones of a different character. Then he reaches another 'node' and switches to the lowest third of the range. The trained singer does not reveal these 'nodes', since he has learned to unify the head, middle and chest register."

A singer who does not understand how registers function will sooner or later have trouble in equalizing his low, middle and high notes. Registers cover each other so that several notes can be sung in both, and there should be no obvious change in going from one to the other.

The term "break" is generally used to indicate the point at which a new register suddenly appears. Whereas the most marked break in the female voice is in its lowest range, the break of the male voice is to be located in its highest range. The majority of writers on the technique of singing advise, for the reasons listed below, that the blending of registers may be achieved by strengthening the upper registers and carrying them downwards.

- (a) It is easier to change from an upper register into a lower than from a lower into an upper.
- (b) There is less risk of straining the registers.

¹Paul Moses, *The Voice of Neuroses*, (New York: Grune and Stratten, 1954), p.92.

- (c) If exercises are practised from below, there is a strong tendency to force a register upwards beyond its proper limits, thus weakening the lower notes of the next register, and greatly increasing the difficulty of a smooth junction.

The incorrect use of the resonating cavities is fatal to the successful blending of the registers. If the resonance rises with the voice, abrupt changes of timbre will be avoided. When descending from an upper register, as much as possible of the resonance of that particular register should be introduced into the upper notes of the register below.

4.3 Falsetto

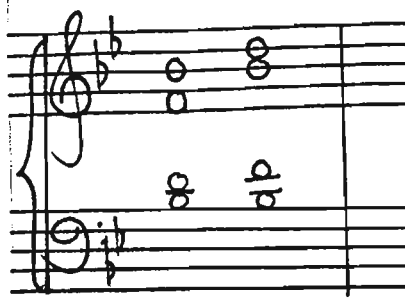
Falsetto occurs when a male singer sings soft notes in the highest regions above his natural range. The usual definitions are, "a false or artificial voice" or, "that part of a man's voice which lies above his natural voice". It was probably first applied to singers in liturgical music in the Roman Church, which did not admit any female singers. The soprano parts were normally sung by boys. During the 16th Century, the polyphonic a capella style had reached such complexity that it was difficult for boys to master it during the relatively short period at their disposal (from about eight to thirteen years of age). Tenors took over the higher parts by singing falsetto, and they were known as tenori falsetti or simply falsetti. Later they were replaced by castrati, i.e. singers whose voices did not break as the result of a surgical operation¹.

During normal singing, the vocal cords vibrate as a whole, whilst in falsetto only the edges vibrate, hence the resulting thin, reedy tone. It also demands the smallest possible column of air. A falsetto note can never be developed into a chest or head note, nor can it be developed to sing very loud. A head note on the other hand, can be sung by singers of both sexes and is performed with partial or complete vibration of the skull and its cavities. A piano or pianissimo note that can be developed to forte is a typical head-note and not a

¹For as long as the Church required the musical services of these singers, the matter was never disclosed, and the Church allegedly denied knowledge about the disfigurement of the boys. Poor and greedy parents were seduced by the wealth that famous castrati could amass.

falsetto. Skilled singers minimize the break between pianissimo head-tones and falsetto by colouring the notes before and after the falsetto notes.

The use of falsetto-singing in choral work is of extreme value. The natural male voice in its high tessitura could very easily dominate the female parts and especially in a final chord such as:



where the F above middle C is initially sung by altos, followed by the tenors which could make it sound pinched or forced. In these cases, falsetto-singing seems to be the logical solution¹.

4.4 Changing Voices

There is a hurdle that all boys meet – the voice change. Boys' voices change drastically, often suddenly and usually with strange, unexpected quirks as they drop as much as two octaves in pitch. Boys then have to learn to sing in a new idiom. The tenor and bass lines are different from anything the boys have had before. There are new harmonic and contrapuntal voice lines, and new clefs.

Extensive voice-tests were undertaken by Frederick Swanson for his Ph.D. dissertation: *Voice Mutation in the Adolescent Male*. He revealed very startling cases where voices dropped in pitch suddenly as much as two octaves, sometimes within the span of six weeks. Of these, some boys also retains a functional treble voice, while others, for a time, could only sing a few deep bass tones and nothing else. Some boys also sang in the low bass and high treble with an area of silence in between, where no sound at all could be produced². Only

¹Philip McLachlan, *Klasonderrig in Musiek*, (Goodwood: Nasou Beperk, 1978), p.133.

²Frederick Swanson, "Changing Voices: Don't leave out the boys", *Music Educators Journal* (January 1984), p.47.

rarely did boys appear who moved down gradually. Swanson also discovered that some boys produced tones an octave below the standard bass clef, also known as the "pulse register" or "Russian bass". These boys are often labeled as deficient singers, but when developed and used properly, they can add an effective depth and richness to any choir.

It is very important for the choral director to be able to recognize the change when it occurs, so that care and attention can be directed towards producing the voice correctly. Some of the symptoms of the changing period are:

- (a) The speaking voice becomes husky and lacks control.
- (b) The speaking voice begins to get heavier in tone and a celestial brilliancy is noticeable during singing.
- (c) The higher notes may disappear or become difficult to sing while the lower notes improve in quality.
- (d) He may experience difficulties in finding the proper pitch and often the following signs are prevalent: raised eyebrows, a wrinkled brow and a pained or "goose-neck" look.

Boys are sometimes taught to use a chest tone for their lower notes before the changing period, as an alternative to using the medium register which may sound weak in comparison to their strong chest tones. The aim in teaching singing is to lay the foundation for a beautiful and resonant voice in adult life. The effort required to force power onto the lower notes of the voice before the change, may easily cause a completely ruined voice. By developing the use of the medium register for low notes, a bridge is established by means of which their voices can gradually pass without difficulty into tenor, baritone, or bass.

It is also of vital importance to use the voice gently during the time of change. The voice will naturally develop in power and improve in

quality if attention is directed to correct breathing and the use of the correct resonance chambers¹.

An adolescent boy should be encouraged to report immediately when he first encounters difficulty in voice-production. From that time on, there should be a constant individual testing and reclassification of the boy's voice until it has settled permanently. Enthusiasm, method, patience and indomitable perseverance on the part of the director is essential.

¹W. Norman Mellalieu, *The Boy's Changing Voice*, (London: Oxford University Press, 1966), p.11.

PART TWO

Chapter 9

The Use of Daily Exercises for Choral Voice Building

9.1 Breathing

Bad methods of breathing cause many vocal "ills", such as tremolo, noisy breathing, etc. Ordinary everyday breathing does not fully inflate the lungs. Correct inhalation and exhalation can only be attained by constant, steady, and patient practice. Avoid practising in a stuffy room and take care not to wear garments that are too tight around the waist as they interfere with rib expansion. Make sure that the body posture is correct, and don't practise too soon after meals as the diaphragm can not act properly when the stomach is enlarged with food. Spasmodic, intermittent practising is of no use whatever. Perseverance over a long period is the only way to ensure success. The majority of choral singers have poor chest expansion at the front and sides of the waist. The following exercises are recommended:

- (a) Lie flat on the back with the head slightly raised. Place one hand on the front of the body at the waist-line and the other at the side. Breathe naturally in and out quietly through the nose. Observe the simultaneous expansion at the two places. Practise for a few minutes until the sensation of using the mechanism in this way is thoroughly grasped.
- (b) Stand against a wall with heels, shoulder blades and the back of the head touching it. Try to obtain the same result as before. Do not take too much breath at first. Practise until certain of the ability to control the mechanism of inhalation.
- (c) Shut the mouth with the lips lightly touching, and inhale through the nose and mouth slowly, silently and steadily. Place the hands lightly beneath the lower ribs, and feel the walls of the chest and the abdomen expand. Exhale suddenly and completely, through the mouth.

- (d) Inhale as in exercise (c). Hold the breath for one second. Without letting the inhaled air escape, try to re-inhale a second time. Exhale immediately. Initially, this exercise will seem difficult, but should eventually greatly increase the air-containing capacities of the lungs.
- (e) Practise inhaling quickly by gradually working at the speed until a quick, light, silent breath can be taken. This exercise must not be done with a jerk or a gasp, and the shoulders must remain very still.
- (f) Inhale slowly, then purse the lips as though about to whistle and let the breath escape slowly, steadily and noiselessly. Counteract the tendency to collapse during exhalation by holding the lower part of the chest and the abdomen well pushed out. Exhale with a steady, unforced, inward and upward pressure of the lower part of the abdomen. The upper part of the chest must never be allowed to cave in, even when the ribs return very slowly to their normal position.
- (g) Repeat the previous exercise, exhaling through an open mouth as in singing. The breath must escape noiselessly, without wavering.
- (h) Inhale quickly, hold the breath for two seconds, and exhale very gradually while supporting the lower ribs and the abdomen from collapsing even more strongly than before.
- (i) As soon as all the foregoing exercises can be performed successfully, the singer may try exhaling while vocalizing on different vowels. Always sing *mesa voce*, in the middle range of the voice. The notes must be steady, with no apparent quivering or breath trembling present.
- (j) Practise as in the previous exercise, but gently swell and then diminish the tone. This exercise will also serve as an introduction to the valuable *mesa di voce*.

9.2 The Tongue

Because the tongue is such a complicated structure, it has often been termed the "unruly" member of the human body. The secret lies in developing the ability to move the tongue (rapidly or slowly) at will, while keeping the jaw perfectly relaxed and motionless.

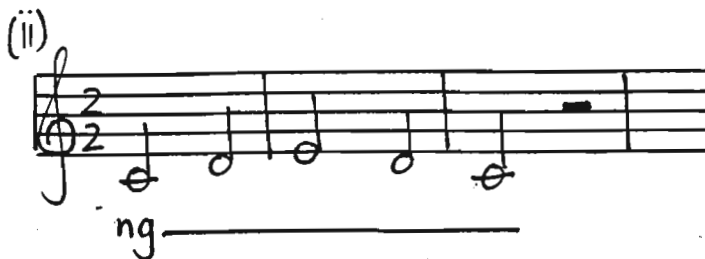
The following exercises should be practised in front of a mirror at least three times per day. Make sure that the lower jaw remains motionless and does not protrude, and that the muscles of the throat do not become stiff.

- (a) Open the mouth wide. Push the tongue out as far as possible and draw it back quickly. Repeat six times without stopping.
- (b) Open the mouth wide. Protrude the tongue as far as possible and move it to and fro, so that it touches both corners of the mouth. Repeat six times without stopping.
- (c) Curl the tongue upwards and backwards as far as possible, so that the tip touches the soft palate. Hold it there for two seconds then return it quickly to its normal position. Repeat this six times without stopping.

The following exercises should induce a suppleness of the soft palate and tongue:

- (a) Drop the jaw loosely, with the tip of the tongue touching the inner side of the lower front teeth. Inhale a short breath through the nose and exhale through the mouth, whispering alternately without voice the syllable "haa" and "ngah". The soft palate and back of the tongue will meet and separate.
- (b) Say "yah - yah - yah . . ." in unison with a relaxed jaw.

- (c) Aim for total freedom of the throat muscles and the root of the tongue while singing as follows^{1,2}:



- (b) Slide up and down on a "r", rolled with the tip of the tongue. Firm breath support makes this exercise possible³.



9.3 Relaxation of the Body

Good singing has the appearance of being effortless. Unfortunately, singing does not become easy by accident. It takes analysis, thought and effort. It also takes hard work, intelligently directed.

¹Exercises (i): Adelheid Armhold, *Singing*, (Cape Town: Tafelberg-Uitgewers, 1963), p. 71.

²Exercise (ii): Wilhelm Ehmann and Frauke Haasemann, *Voice Building for Choirs*, (London: Barenreiter Verlag, 1982), p.30.

³ibid.

A correct emission of the voice is always accompanied by freedom of the face and body. The eyes, especially, reveal a lot as a good tone will always be accompanied by a soft and relaxed expression in the eyes.

- (a) Devote at least ten to fifteen minutes daily lying on the back in a completely horizontal position. Try to empty the mind of all thoughts, then slowly will a feeling of relaxation over every part of the body until it reaches a point of complete slackness. The gradual releasing of tensions should bring about a sensation of extreme relief.
- (b) Stand erect, drop the head forward, then roll the head in slow circles while keeping as relaxed as possible .
- (c) Move the shoulders up and down in slow circling movements.
- (d) The following exercises are recommended for loosening the jaw:
 - (i) Whisper without voice the syllables: "Fah", "Thaw", "Mah" and "Wah". Raise the jaw while pronouncing the consonants and drop it loosely downwards for the vowel.
 - (ii) Simply drop the jaw as if one is falling asleep.
- (e) Physical activities should be encouraged because they produce the flexibility and relaxation needed for singing. Suggested forms of exercise are: swimming, jogging, ball games and stretching exercises. Through these forms of exercise and relaxation, body awareness is heightened, preparing the way for body-mind co-ordination.

Chapter 10

Voice Building

10.1 Voice Warming Exercises

The singer, like an athlete, has to go through a "warming-up" process prior to putting forth his best efforts. The muscles should be gradually limbered up into a state of readiness and full efficiency for what lies ahead. It is the author's experience that "warming-up" should form an integral part of any rehearsal. It is recommended strongly that choral conductors should lead his choir members through the following "warming-up" sequence:

- (a) Stand up, assuming a good posture. Make sure that the head, neck, shoulders, arms, throat and jaw is relaxed.
- (b) Inhale and exhale several times, until the breathing mechanism comes under conscious control.
- (c) Gradually ascend and descend in semitones while singing the following exercises^{1,2,3}. Choose a tempo which is comfortable for the choir.



¹Exercises (i) to (v): Philip McLachlan, *Klasonderrig in Musiek*, (Goodwood: Nasou Beperk, 1978), pp.127-128.

²Exercises (vi) to (viii): Viktor Fuchs, *The Art of Singing and Voice Technique*, (London: Calder and Boyars, Ltd., 1963; reprint ed. 1967), p.117.

³Exercise (ix): James Lawson, *Full Throated Ease*, (New York: Mills Music, Inc., 1955), p.62.

(ii)

muu muu non non whe whe claw claw etc.

(iii)

wha wha ——— wha muu muu muu ——— muu

(iv)

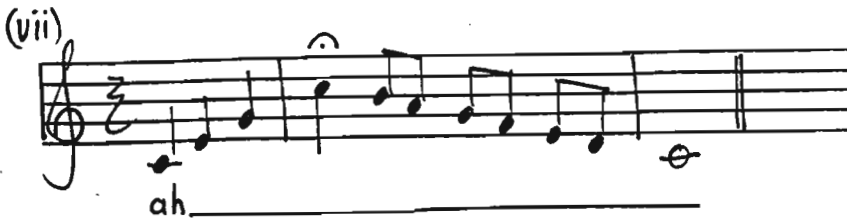
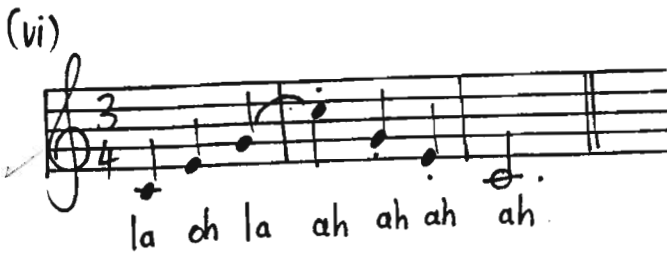
yo—ong-e yo-ong-e yo-ong-e yo-ong-e yo—nge

(v)

ha_ ha ha ha ha _ _ _ _ _ ha

haw_ haw haw haw _ _ _ _ _ haw

hee_ hee hee hee _ _ _ _ _ hee



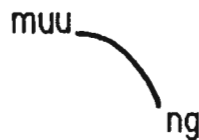
10.2 Resonance

Good resonance is the key to good tone. In order to develop this technique, a choir should devote about five minutes of each practice session to strengthen resonance. Begin all resonance exercises by singing piano and aim to develop a feeling of emergence of sound from

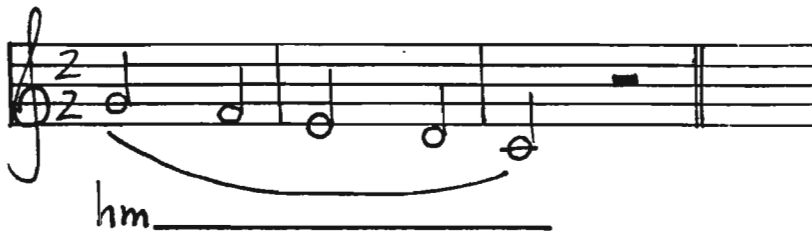
within the body. The facial muscles must be very relaxed. Useful exercises to encourage good resonance are the following:

- (a) Relax the jaw and rest the lips gently on each other as if they were thick. Allow as much room in the mouth as possible by parting the teeth slightly. To encourage forward placement of the sound, shape the mouth slightly in a fishmouth form. Place the hands in front of the mouth and hum into them. Glide from the vowel sound to a sustained consonant:

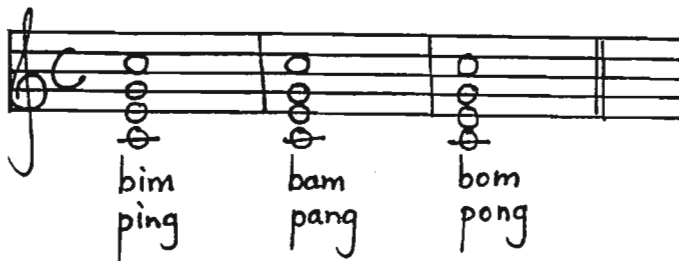
muu
ng



Advance to fixed pitches, for example¹:



- (b) Resonance exercises are very successful when applied to a chord. The chords should be transposed by conjunct interval steps².



¹Wilhelm Ehmann and Frauke Haasemann, *Voice Building for Choirs*, (London: Bärenreiter Verlag, 1982), p.33.

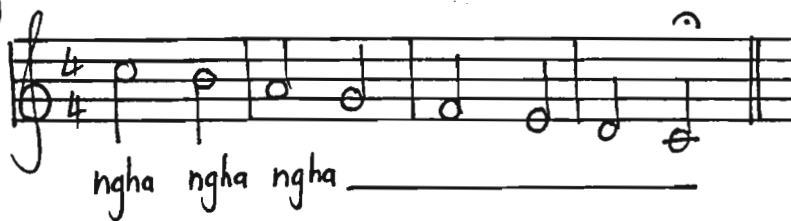
²Ibid., p. 34.

- (c) Sing the following exercises while gradually ascending and descending in semitones^{1,2}:

(i)



(ii)



(iii)



(iv)



¹Exercises (i), (vi) and (vii): Ibid., p.34.

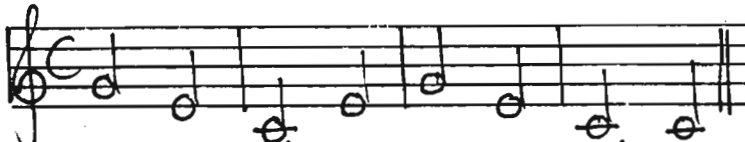
²Exercises (ii) to (v): Norris Croker, *Handbook for Singers*, (London: Augener, Ltd., n.d.), pp. 29, 116-117.

(v)



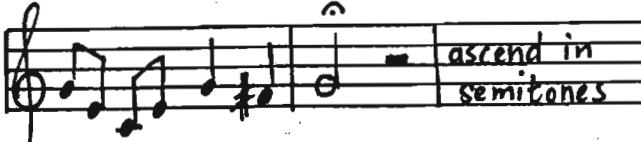
mm-ah, mm-ah mm-ah, mm-ah
 mm-eh, mm-eh mm-eh, mm-eh
 mm-ee, mm-ee mm-ee, mm-ee
 mm-oh, mm-oh mm-oh, mm-oh
 mm-oo, mm-oo mm-oo, mm-oo

(vi)



mam-ma mi-a mam-ma mi-a
 much more mo-hey much more mo-hey

(vii)



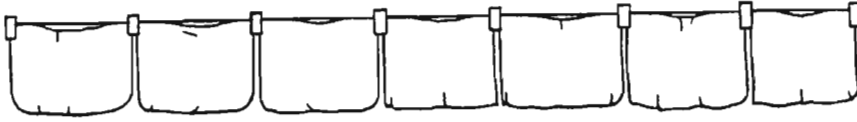
mi-na mi-na mi-na
 ascend in semitones

10.3 Clear Articulation

The correct application of words to notes requires a lot of patient care. Vowels open the throat, consonants close it. So we have to learn to articulate consonants in such a way as not to impede the flow of the sound, as a beautiful tone is the foundation of good singing. That is why the Italian language, with its pure vowels and lightly articulated consonants, serves the purpose of freeing the throat and developing flexibility of the tongue and lips admirably.

Practise initially on the vowel syllables only, until they blend perfectly without cessation of sound. When maximum freedom of the throat has been attained during the changing vowel process, the con-

sonants can be added. They must be sung as fast, light, and distinct as possible, helped by the lips and tongue, while the jaw drops loosely. They should be like the clothes-pins that secure the sheets of vowel sounds on the line, yet separate one from the other¹:



Exercises on articulation and pronunciation of vowels.

- (a) Inhale, drop the jaw, and let the tongue assume the position for "ah". Whisper on one continuous outflowing breath the syllables "ah-ee ya-ee-ya". Make sure to glide from one vowel into the other. The tip of the tongue should remain in contact with the lower front teeth, and the back part of the tongue in a low position. The action of the tongue must be loose and supple.
- (b) In the same way, whisper the vowel "ah" in conjunction with each of the following vowels²:

Bright vowels:

ee as in him -	ah ee ya ee ya
e as in men -	ah e ya e y
ai as in main -	ah ai ya ai ya
a as in man -	ah a ah ah ah
e as in her -	ah e ah e ah
uh as in up -	ah uh ah uh ah

Dark vowels:

o as in on -	ah o ah o ah
aw as in saw -	ah aw ah aw ah
o as in go -	ah oh ah oh ah
oo as in boo-	ah oo ah oo ah
oo as in soon -	ah oo ah oo ah

¹James Lawson, *Full Throated Ease*, (New York: Mills Music, Inc., 1955), p.43.

²George Dodds, *Voice Placing and Training Exercises*, (London: Oxford University Press, 1927), p. 34.

As soon as these vowels can be whispered correctly and with facility, they should be sung on notes lying in the middle part of the voice. The jaw should hang loosely to avoid stiffness.

- (c) Whisper the following consonants, combined with vowels, very clearly and crisply¹:

Lah lay lee lo loo
 Dah day dee doh doo
 May may mee mo moo
 Vah vay vee vo voo
 Quiet quaint quest quit quote
 Rah ray ree ro roo
 Nah nay nee no noo
 Bah bay bee bo boo
 Thah they thee tho thoo
 Cries craze freeze throws bruise

- (d) Double consonants can be practised by using the following list of words²:

Up past
 If felt
 His zeal
 Good day
 Each chap
 Will Live
 Dumb brute
 I've vowed
 With thee
 Take care
 Large jar
 In next
 Some more
 Eyes smile
 Don't tell

¹Ibid., p. 34.

²Ibid., p. 35.

Big game
Push sharp
Purest star

(e) Practise the following for clear enunciation¹:

In next	In ecstasy
A nymph	An imp
Attend	At end
Offend	Off end
Askew	Ask you
A nice house	An ice house
The cuckoo sings	The cook who sings

(f) Say the following tongue-twisters as quickly as possible²:

"She sells sea-shells on the sea-shore. The shells she sells on the sea-shore are sea-shells, I'm sure."

"Around the rough and rugged road the ragged rascal ran their rural races."

"Never knit knowingly knotted knubs of knitted nothing."

"If Peter Piper picked a peck of pickled peppers, how many peppers did Peter Piper pick?"

"The bleak breeze blightens the bright blossoms."

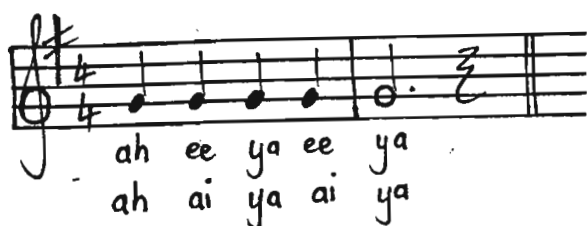
"Flesh of freshly fried flying fish."

¹Ibid., p. 35.

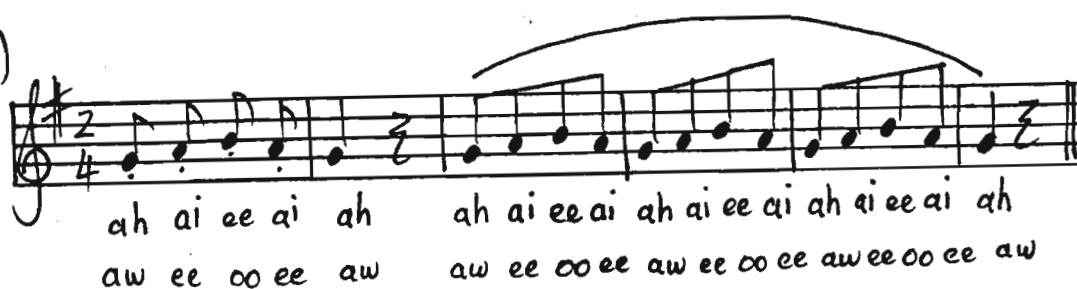
²Adelheid Armhold, *Singing*, (Cape Town: Tafelberg-Uitgewers, 1963), p. 76.

- (g) Sing the following suitable vocal exercises while ascending and descending in semitones^{1,2,3}:

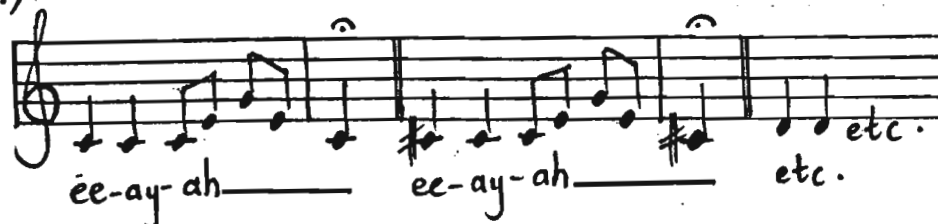
(i)



(ii)



(iii)



¹Exercises (i) to (iii): Ivor Warren, *The Examination Candidate's Handbook on Singing*, (London: A. Hammond & Co., n.d.), pp. 17-18.

²Exercise (iv): Adelheid Armhold, *Singing*, (Cape Town: Tafelberg-Uitgewers, 1963), p. 75.

³Exercises (v) to (vii): George Dodds, *Voice Placing and Training Exercises*, (London: Oxford University Press, 1927), p. 34.

(iv)



(v)



daw	daw	_____
flee	flee	_____
grü	grü	_____
bra	bra	_____
fra	fra	_____

(vi)



(vii)



(viii)



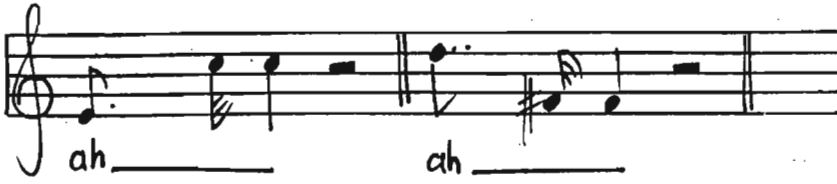
10.4 Legato Singing and Slurring

Legato singing implies that there is no cessation of tone between a note and its successor. In this way, the whole phrase is bound together (unless specially marked to the contrary) in one continual stream of tone. In this method of singing, the singer has the opportunity to display good phrasing, and fill the music with a volume of beautiful sound.

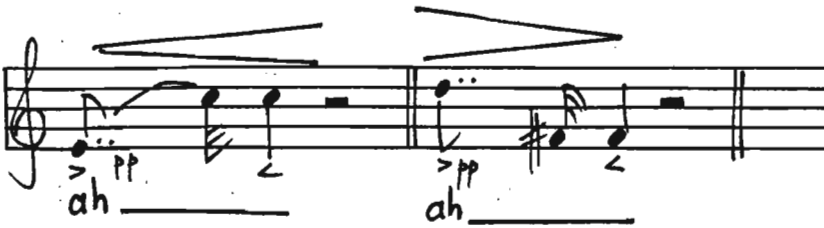
To slur is to carry the voice, either quickly or slowly, from one note to another. It is an ornament that should only be occasionally employed for special effects, as its frequent use leads to a sickly sentimentality or even scooping. The intervening notes are only heard indistinctly and should not sound as if the voice sings a scale, but rather be compared to a violinist sliding his finger up or down the string. The general rules regarding slurring are:

- (a) The slur must occur in the time of the note slurred from, not of that slurred to.
- (b) When ascending, a slur is usually done with a slight crescendo; when descending, with a slight diminuendo.
- (c) The slur should never be used several times in succession or between two notes of short duration.
- (d) The slur should be very seldom used when the interval is small.

In order to acquire the correct management of the slur, interval singing should be practised without any slur, but slightly anticipating the second note¹:



As soon as this can be done correctly, practice the slur ascending and descending from one note to the next. Start slowly, and gradually increase the speed of the slur. Always slightly anticipate the note slurred to, as in the above example. In order to do this properly, a diminuendo should be made on the first note, the slur itself should be sung very softly, with a crescendo starting as soon as the second note is reached²:



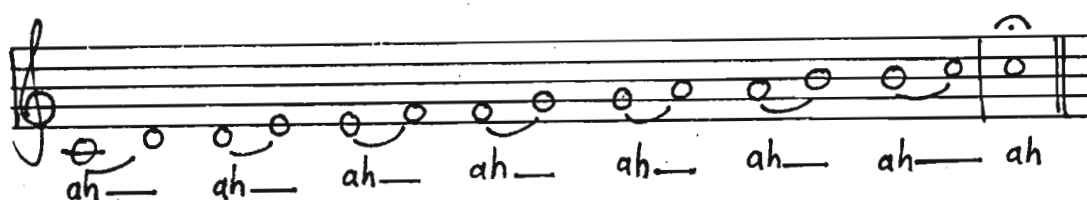
A slur becomes a serious fault when it is constantly used. Try to correct this by practising legato singing. The main differences between slurring and legato singing is:

- (a) The intervening notes in a slur are heard while legato is a very rapid slur where no intervening notes are heard.
- (b) The time of the note slurred from is robbed of some of its value while in legato singing no time is taken from any note.

¹Ibid., p. 90.

²Ibid., p. 90.

In order to acquire legato, practising should be as for the slur, but without anticipating the second note. The following exercise¹ should be sung as smoothly as possible:



10.5 Flexibility and Agility

These two terms are often confused as having the same meaning. They actually imply two very different accomplishments.

Flexibility means the ability to graduate the power of tone from *pp* to *ff*, as well as illustrate and heighten the atmosphere of the song by adding the proper tone colour. Agility means the ability of the voice to move easily and rapidly from one note to another, whatever the interval or speed may be. An agile voice is a freely produced one, with complete absence of strain and tension. The difference between the two is that flexibility is mainly a matter of aesthetic taste, while agility is a mechanical process.

¹Judith Litante, *A Natural Approach to Singing*, (London: Oxford University Press, 1962), p. 44.

The following exercises^{1,2,3,4} are all very useful for improving agility. Generally, these exercises are also of considerable help in extending the range of the voice. As the pitch rises the lower jaw should be totally relaxed, and should drop sufficiently to allow for the appropriate mouth opening. A rapid increase in power towards the higher notes should be avoided.



¹Exercises (i) and (ii): Ivor Warren, *The Examination Candidate's Handbook on Singing*, (London: A. Hammond & Co., n.d.), pp. 18–19.

²Exercises (iii) to (v): Adelheid Armhold, *Singing*, (Cape Town: Tafelberg-Uitgewers, 1963), p. 71.

³Exercise (vi): Judith Litante, *A Natural Approach to Singing*, (London: Oxford University Press, 1962), p. 47.

⁴Exercise (vii): Mathilde Marchesi, *Vocal Method*, (London: Enoch & Sons, n.d.), p. 34.

(iii)



(iv)

1. forte
2. piano

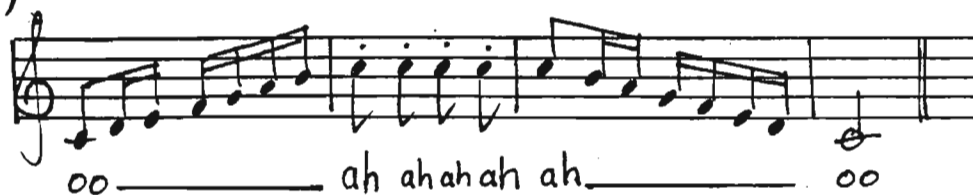


(v)

1. forte
2. piano



(vi)



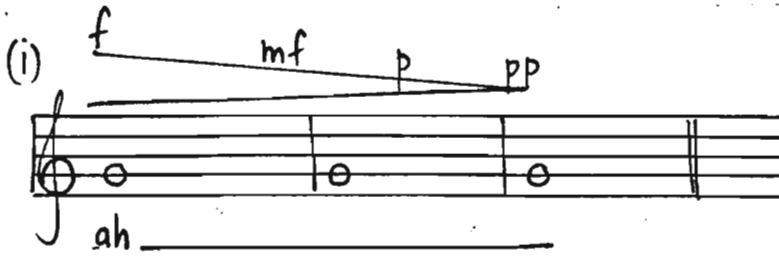


Messa di Voce is a gradual crescendo from the softest to loudest tone, followed by a gradual diminuendo back to the power of initial tone. It requires absolute calm and relaxation. Visualize a firm, open "ah" at the back of the throat and sing the swell by mentally willing the tone to swell and fade. The upper part of the chest must be held out firmly and the control of the abdominal muscles must be even more strenuously exerted than in ordinary singing. The control of the breath is the secret of success in this ornament.

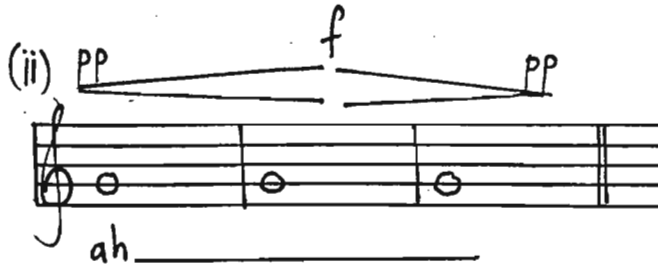
The following exercises¹ are recommended for developing "messa di voce":

¹Ivor Warren, *The Examination Candidate's Handbook on Singing*, (London: A. Hammond & Co., n.d.), p. 19.

- (a) Attack firmly and gradually diminish the tone by decreasing the breath pressure. Try to maintain the same quality of tone to the end.



- (b) Attack softly, gradually swell out and decrease the tone gradually again.



10.6 Covering and Colouring Tone

Covering and colouring often get confused because opinions amongst performers and teachers vary greatly. Covering and colouring are both ways of refining tone.

Covering is a technical device which helps to give greater intensity and beauty to the transitional notes, which are:



in sopranos and tenors, and



in baritones and basses.

Physiologically, covering is a quality that a singer gets when his larynx rests in a low position. This position creates another whole resonance tract that gives the vowel the typical "dark" or covered tone. During an ascending scale, a singer must modify the vowel imperceptibly or the vocal line will be distorted. This slight vowel alteration is known as covering the tone.

One could say that each voice has its own individual colouring. For example, there are various voices with a definite silvery quality. This individual colouring in an interpretative sense, is an effect produced by means of the emotions of the singer. This emotion must never be of such a nature as to destroy the singing. Like an actor, the singer should feel and participate, but there must be a degree of controlled detachment. There are also two purely technical means of colouring the voice. Firstly, the "singer's smile" may be used to brighten the tone. Secondly, the singer may deliberately increase the extent of nasal resonance which serves to enhance the "warmth" of the tone.

Chapter 11

Voice Training Dealing with Specific Technical Problems

11.1 Forced Tone

Undue tension will cause muscles to tremble. We can compare it with lifting weights beyond our strength where our muscles tend to tremble after they are relieved of the weight. So is it with the vocal muscles as well, which will develop a tremolo from forcing. Pushing, shouting, coughing and screaming are all enemies of the vocal cords. Singing depends entirely on two tiny membranes, which once overstrained, like two rubber bands, can never regain their former elasticity. High notes especially, need more muscular support. If the breath is not prepared in time, the note will be sung with force. The following exercises are designed to cure the problem:

- (a) General relaxation exercises¹. Calm, relaxed (but concentrated) practice will restore the ability to sustain vowels without quivering.
- (b) The cure for vocal nodules is to stop talking and singing completely. If the growth is very big, only an operation can help. The operation is difficult and dangerous, as the slightest injury to one of the chords may permanently damage it.

11.2 Frontal Tone

The tone may be described as cold, colourless, monotonous and unimpressive. It is caused in two ways:

- (a) By tightening and stiffening the muscles of the throat, especially when attacking high notes.
- (b) By projecting the tone too far forward, so that it escapes before it is amplified by the action of the resonance activities.

¹Refer to exercises in part 2, p. 61.

When the muscles of the throat start to tighten and harden, sing the phrase again, and try to keep the muscles of the throat supple and relaxed by exercising sheer will power. Humming the phrase should also help. Test the relaxation of the throat by turning the head slowly from left to right while singing the note or phrase on which the stiffening of the muscles occur. The improvement in tone should be heard at once.

11.3 Guttural or Throaty Tone

This is caused by the root of the tongue being stiffened or raised, and pushed backward simultaneously until it almost touches the pharynx. The breath then has to be violently forced past this obstacle resulting in great strain on the windpipe. The following exercises are designed to cure the problem:

- (a) Exercise will power to keep the root of the tongue relaxed and flat in the mouth while sounding the vowel, or vowels, which cause a throaty tone.
- (b) While sounding the vowel Ee as in peep (the vowel on which a guttural sound is the least likely to occur), try to keep the back of the tongue from rising¹.



- (c) If the tongue is still obstinate, refer to part two, p. 59 for further exercises.

¹Walter Ehret, *The Choral Conductor's Handbook*, (New York: Edward B. Marks Music Corporation, 1959), p. 38.

11.4 Husky Tone

This is either due to:

- (a) a bad cold; in which case one should rest until the cold is over.
- (b) an over-strained voice. This is a frequent problem and is a symptom of incorrect vocal production. The voice should be rested for a few weeks. The choral director should then impress upon the singer the correct techniques of voice projection as discussed in this part of the thesis.

11.5 Inability to Sing Softly

The basic problem may be too much muscular pressure on the voice. Too much breath is taken too often, causing the vocal mechanism to be in a state of tension. The following exercises are designed to cure the problem:

- (a) The mind, not the throat, must propel the voice, especially in soft singing. A soft tone is an exact replica of a loud one, but with less intensity of sound. It involves no change of mechanism. The position and colour must remain the same, while the throat remains utterly passive. Experiment with a phrase spoken loudly, then softly, and try to reproduce the same natural responses when singing. There is an old Italian saying, "Seek for quality. Quantity will come".
- (b) Sighing and yawning will relax the singer as well. Breath should be taken conservatively and care should be taken that the singing remains at a relaxed dynamic level.
- (c) Songs with echoes can be sung or the conductor can play or sing a phrase loudly and ask the choir to sing it back as an echo.

11.6 Loss of Voice and Fatigue

Hoarseness or total loss of the voice does not always imply a state of infection, but are often due to fatigue or severe emotional stress. The larynx reacts to stress just as many other parts of the body do.

Emotional problems tighten the vocal cords, and this may, in turn, cause the volume and pitch of the sound waves to change. Total loss of voice may even be the result of a perfectly normal larynx paralysed by acute anxiety.

A sore throat is the most important alarm signal in the body, and the singer should be excused from all practises or performances and sent home. A singer should also refrain from singing, talking and laughing too loudly when he is ill. When such risks are taken, the results can be very costly, or even devastating. The reason the singer damages his voice is that, in order to compensate for the effects of the sore throat, he begins to use incorrect vocal techniques.

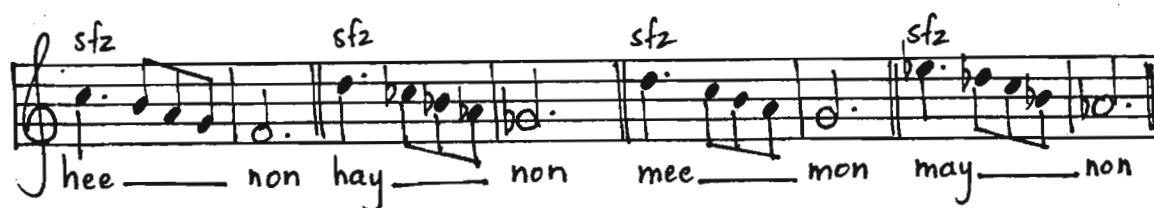
The vocal cords should be examined by a specialist to check that nothing is physically wrong. The only real remedy is to rest the voice for as long as the condition requires.

11.7 Nasal Tone

When the soft palate and tongue drops too low, the exit of tone through the mouth is partially shut off. Should the sound stay locked in the nose, worsened by a small mouth space, the tone is decidedly nasal. It must be pointed out that nasal tone and nasal resonance are two totally different effects. The raising of the soft palate will not interfere with the proper use of the nasal cavities for resonance. Practise the following exercises in front of a mirror:

- (a) Open the mouth well and make sure that the root of the tongue is lying flat in the mouth. Draw in a deep breath through the nose, then half-way through, inhale forcibly through the mouth. While inhaling through the nose the soft palate is low, but when the method of inhaling is changed to a forcible inspiration through the mouth, the soft palate flies upwards. Sing the vowel tones on the notes which are nasal, taking great care that the soft palate never drops to a very low position.

- (b) Try to expand and contract the pharynx. Draw the uvula backwards and upwards. These feats may seem difficult initially, but the control gained will be invaluable. The following exercise may help to alleviate the problem¹:



11.8 Palatal Tone

This problem occurs mainly in the higher range of the voice, and is characterized by a throttled, nasal tone. It is caused in two ways:

- By curving the centre of the tongue upwards, resulting in the tone striking the soft palate instead of the hard palate.
- By the pharynx being insufficiently expanded which robs the tone of its proper vibration.

The following exercises are designed to cure the problem:

- It will be noticed that in sounding the Ee vowel the tongue arches upwards just behind the tip. Practise the tongue exercises².
- Hum the intervals of an ascending scale. Open the mouth to sound the vowel-tone Ah, as in "lark", as soon as the faulty tone is reached. Do not breathe again and keep the pharynx expanded throughout the exercise.

¹Dante J. Smit, "Sekondere Kooronderrig", (M. Mus. thesis, College of Music, University of Cape Town, 1983), p. 41.

²Refer to exercises in part 2, p. 59.

11.9 Poor Intonation

The single most important cause of poor intonation is lack of discipline amongst the choir members in following the rules regarding correct body posture and breathing methods¹.

The following are other reasons for poor intonation. Appropriate suggestions to correct the problems are given:

- (a) Not enough extra effort is put into an ascending scale or interval, resulting in a flat tone. Likewise, the exaggerated relaxation which occurs when singing a descending interval also results in a flat tone. The cure seems to be to think of "large" intervals when the music ascends and exactly the opposite during descending passages. Roe² suggests the following mental picture, "Approach ascending intervals like walking upstairs in total darkness, and descending intervals like walking downstairs in the dark".
- (b) When singers have to sing high notes on difficult vowels, voice strain and faulty intonation may result. Singers should be asked to open their mouths as wide and relaxed as possible. The result will be an altered vowel sound which is quite acceptable to the ear.
- (c) One or more of the voice groups may have to sustain a note for several measures. As soon as they run out of breath a flattening of the tone occurs. It may help to stagger the breathing of that particular voice group or to insert crescendos and diminuendos corresponding to the natural rise and fall of the music. It may also help to mentally pitch the note repeatedly on each beat of the bar. The intensity level should always be kept up by feeling the "drive" and "follow through" while holding the long notes. Likewise, this "forward movement" must be kept in mind when

¹Refer to breathing exercises in part 2, pp. 57-58.

²Paul F. Roe, *Choral Music Education*, (Englewood Cliffs: Prentice-Hall, Inc., 1970), p. 120.

singing slow tempi. The tone must be kept supported, vibrant and alive throughout its duration.

- (d) Numerous repetitions of an a capella selection may tend to become progressively flatter. To snap the choir out of the "worn musical groove" and sharpen the concentration, the pitch should be raised by a semitone. If the tessitura of one or more of the voice groups centres around an uncomfortable area of the voice, the music may be transposed a semitone up or down, as required, to ease the strain. The conductor should take care not to group two numbers in the same key, or related keys, together in a programme, because tonal deafness may occur when no key relief is provided¹.
- (e) Singers often use strained chest tones when singing low notes. The head register should be mixed with the low chest tones which should result in an improved quality of sound.
- (f) Excessive tension and strain on the vocal cords result in a sharpening of the pitch. When the strain is caused by a particular passage being too high, a modification of the vowel sounds may be helpful. Singers should realize that forcing the power, range and registers of their voice is self-defeating. A conductor using tense arm muscles will also cause singers to strain and "push" their vocal chords. Instead, a "hushing up" gesture from the choir director can be of assistance.
- (g) Conductors can be the cause of bad intonation by their insecure cueing. For example, an insecure cue to the sopranos may result in their being inadequate prepared to take a sufficient breath to last through a long phrase. The conductor should memorize the music and practice the appropriate conducting movements for important cues in front of a mirror.
- (h) A lack of tact and sensitivity on the choral director's part can cause anxiety amongst the choir members. An anxious singer can

¹Imogen Holst, *Conducting a Choir*, (London: Oxford University Press, 1973), p. 4.

not breathe properly, and a singer with inadequate breath control can not sing in tune¹.

- (i) Physical and mental fatigue, particularly during examination periods and before a competition or big production, should be expected. The conductor should follow a well organized and efficient rehearsal programme with specific weekly objectives in order to remove the necessity for strenuous, last minute, extra rehearsals.

The conductor should introduce variety to enliven the rehearsals. For example he may beat erratically or have the choir stand and stretch, sing the number staccato-style on the words or do a few deep-breathing exercises.

- (j) Excessive heat, humidity or poor ventilation has a depressive effect on singers. Instead of feeling perky and vital, they feel emotionally, physically and vocally drained, which soon leads to flat singing. The director must then use extra vitality in his direction, make sure the singers breathe deeply and that their posture is correct.
- (k) Sliding, scooping or approaching notes from below, and even anticipating the pitch of the next note before the actual beat on which it is to be sung can be eliminated by insistence of a mental and physical approach from above. Movement from note to note must be made with "square" corners. Another device is to practice the voice parts staccato or marcato.
- (l) Soft passages are often sung flat because most singers lack the mental and physical intensity to give the tone enough vibrancy. The singers should sing their tones loudly and immediately repeat it softly while retaining their identical mental and physical sensations. Extra support is needed to correct flat and breathy soft singing.

¹Ibid., p. 5.

- (m) There is a tendency, when singing repeated notes, to progressively drop the pitch. The choir should be told to think of each note as being higher than the one preceding it. McLachlan¹ suggests that a repeated note be approached from above, almost as if an acciaccatura is added:



- (n) A poor time of the day for rehearsals, especially early mornings, is inadvisable. The voices are not yet "warmed up" and this may easily lead to poor intonation and may even induce strain on the voice. It is advisable to follow a systematic rehearsal plan and warm up the vocal apparatus, with carefully planned exercises.
- (o) Poor acoustics on stage which prevent members of the group from hearing each other, may easily lead to distress, especially when polyphonic music is performed. Too much echo in the venue, caused by badly positioned reflective surfaces, such as windows and hard walls, will cause a sharpening of the pitch. Conductors should compensate in overly-live rooms by avoiding tempi which are too fast. Conversely, the deadening effect caused by excessive absorption of sound by large audiences, and such things as curtains and rugs, will tend to cause a drop in pitch. This tendency can be countered by choosing a slightly faster tempo.
- (p) Any accidental calls for an even greater change of pitch than the sharp or flat would represent in the key signature. The voice must, therefore, pitch higher for the sharp sign and lower for the flat sign than would normally seem natural in the original key. The choir members will easily sing sharp if they do not deliberately pitch a tone lower when a flattened accidental occurs, especially the third of a minor chord and the diminished

¹ Philip McLachlan, *Klassonderrig in Musiek*, (Goodwood: Nasou Beperk, 1978), p. 142.

seventh of any chord. Conversely, singers tend to sing flat if the third of any major chord is not pitched high enough; if the leading note of a scale is pitched very high; if the sixth is not slightly raised; and if the major second is not sung as a big step¹.

11.10 Register Disorders

When there is a very sudden change of quality in passing from one so-called register to another, it is known as a "break". It is often caused by carrying a lower register too high (very often, the chest register in female voices) which results in a sudden violent change in the mechanism of production. The aim should be a complete unity of colour, which would require an undeviating vowel. Think of linking one tone to the other like an indissoluble chain².



At each point of amalgamation there are several notes that can be sung in either one of two registers. As a result registers overlap. Except for special effects, the higher rather than the lower register must be used. There are two reasons for this choice of register. Firstly, if the lower register is selected, the tone will sound harsh, or "open" as it is properly known. Secondly, strain on the vocal cords is avoided.

¹For an interesting account of the evolution of the scale, refer to Paul Roe, *Choral Music Education*, (Englewood-Cliffs: Prentice-Hall, Inc., 1970), pp.117-119.

²Judith Litante, *A Natural Approach to Singing*, (London: Oxford University Press, 1962), p. 38.

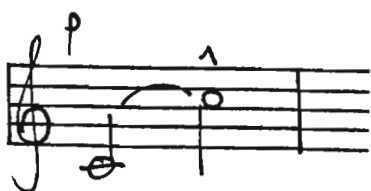
The following exercises¹ are designed to assist the singer to amalgamate his low and middle registers:

- (a) Sing a descending scale of 5 notes, commencing well up in the middle register. Transpose the scale a semitone lower each time until the voice is carried well below the point of amalgamation.



Sing with a full soft tone, forcing neither the power nor the breath. Ensure that the quality of the starting note is good and try to maintain the quality, power and tone, as well as the purity of the initial vowel sound throughout the exercise. Avoid any alteration of the position of the larynx, and all the other movable parts inside the mouth. Make full use of the resonating cavities. Do not attempt the second exercise until this scale can be sung with no alteration in the quality from the first to the last tone.

- (b) Sing an interval of an octave:



Start on a note in the low register, using the vowel tone as if to sing in a higher register. Carry the voice upwards to the octave above, striking the higher note decisively, but not too loudly. Place the fingers lightly on the larynx to ensure that it does not rise. The two notes should sound alike in quality. If the higher

¹Norris Croker, *Handbook for Singers*, (London: Augener, Ltd., n.d.), p. 29.

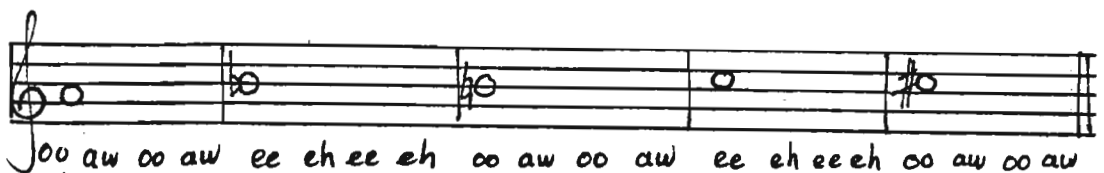
tone sounds too open, the wrong register is being used. The sensation to be felt should be as though the higher note were dropped down from above, not approached from below.

- (c) This is a combination of the two previous exercises. Sing an interval of an octave followed by a descending scale of five notes. Transpose the exercise down by a semitone each time and sing it in one breath.

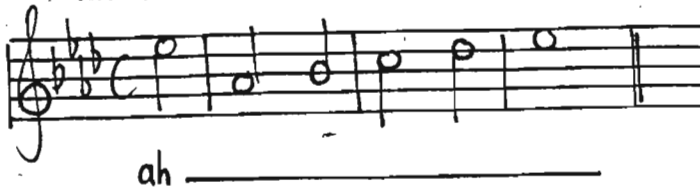


In order to amalgamate the middle and high registers the following exercises are prescribed:

- (a) Sing an ascending scale of five semitones and ensure that the quality of the starting note is good. The whole exercise must be sung with a soft, round tone and the larynx must be kept low. Aim to make the quality of the two registers sound as much alike as possible. No force must be used, so that the quality and power may develop naturally. If the voice is carried too high, thereby forcing the compass, the voice will be ruined.



- (b) In this exercise, start on a note well above the point of amalgamation. Transpose the starting point a semitone lower each time. Keep the directions of the previous exercise in mind.



Finally, to equalize all three registers the singer can use this exercise:

Start a scale on the lowest note in the compass, and sing slowly upwards to the octave above, and descend slowly again. Transpose the exercise a semitone higher at each repetition. Maintain the breath pressure equally throughout the exercise and sing the high notes by altering the tension of the vocal cords and adjusting the voice according to the various registers in use. The same strength of tone must be maintained from the first to the last note. The larynx must be kept low, the pharynx expanded, and full use must be made of the resonating cavities. It must be remembered that increased power is only gained by increased force of breath emission and even greater use being made of the resonating cavities. Support very well, especially during mezzo-forte and forte singing.

If any notes in the high register are badly produced, hum a scale up to that note, and without breathing again, open the mouth and sing on the given vowel-tone. Take care that the vibrations are not felt in the bridge of the nose itself, as the tone will then be nasal.

The muscles of the throat must not stiffen at all during the practising of these exercises, or the tone will suffer. Check this by placing the fingers very lightly on the throat between the larynx and the jaw, and hum the exercises immediately if the muscles are found to stiffen in the slightest.

11.11 Shock of the Glottis

Shock of the glottis is liable to occur when a vowel is sung after a period of silence. Should the singer force his vocal cords into vibrating immediately the larynx is opened to allow the breath to pass, an ugly tone results. This brutal or explosive attack on the vocal cords results in poor resonance. It can be avoided by allowing the breath to start flowing before the inception of the sound.

In preparing to sing, the mind conceives of the tone to be uttered, which, in turn induces a reflex action of the muscles and nerves. They regulate the necessary amount of breath to be taken as well as the correct tension and vibration of the vocal cords. By mentally singing a "h" before the initial vowel, the correct contact between the vocal cords and the flow of breath is ensured.

11.12 Shrill Tone

High notes are often shrill and harsh because of the following reasons:

- (a) The position of the mouth and jaw remains fixed throughout the range.
- (b) Notes lying in the high register may induce the feeling of climbing up a dangerous ladder. This incorrect conception causes the larynx to rise with the result that the singer clings desperately to the resonances behind the nose where there is not nearly enough room to amplify these notes.
- (c) There is not enough breath support¹.

The correct position for high notes is a relaxed jaw, with the mouth opening twice as wide as for vowels on lower notes. This position causes the larynx to drop considerably taking the root of the tongue with it. The higher the note, the wider the cheeks stretch sideways. A so-called "high" note is merely a sound produced with a smaller

¹Refer to breathing exercises in part 2, pp. 57-58.

opening between the chords. This results in more tension and consequently a faster vibration of the cords. It is important to remember that sound still originates in a low place, deep in the throat. High notes find their resonance along the back of the head as well as sympathetic resonances in the top part of the chest.

For problems (a) and (b) above use previous exercises¹ on the "oo" and "ah" vowels. Ascend chromatically while opening the resonators to the sound. Relax the tongue and jaw.

For problem (c) use the breathing exercises.

11.13 Tone-Deafness or Monotism

The characteristic of a tone-deaf person is a relative deficiency in tonal memory and pitch discrimination. He either sings out of tune persistently, or growls (drones) on one note or a few notes only. Monotism in most young people appears to be a form of retarded musical development, or a lack of the appropriate conditioning of reflexes. As a rule these people do not reveal any significant deficiency in rhythmic memory.

While some children develop the ability to remember the "essential" ingredients of the tune in detail at a very early age, others do not achieve it even by the age of eleven or twelve years. A few people never achieve it at all. These are the children and adults referred to as monotones. According to an experimental study in tone-deafness by D. Fry² during 1948 in England, he found that 5 per cent of the population are likely to be tone deaf. Symptoms of this condition include:

(a) A limited range.

(b) Poor vocal quality.

¹Refer to the exercises prescribed for the amalgamation of middle and high registers in part 2, pp. 92-93.

²D. Fry, "Tone Deafness", in *Monotones*, ed. Arnold Bently (London: Novello & Company Ltd., 1968), p. 19.

- (c) Difficulty when using the head voice.
- (d) Lack of confidence in the ability to sing.
- (e) Lack of attention, resulting in slowness when learning a song.
- (f) Aloofness or even hostility towards music, which occurs even more amongst boys.
- (g) Poor musical memory.

If a cure is possible then a psychological approach to the problem should be followed.

- (a) Persuade the child that he can sing.
- (b) Patience, encouragement and imaginative exercises on an individual basis may help eventually. Exercises should initially be based on tune- and pitch discrimination. Thereafter short and concentrated tonal and rhythmic memory tests should be introduced with attention paid to correct vocal production. Music with a familiar tune especially, will encourage the central nervous system to carry out responses eagerly.

11.14 Tremolo (Wobble) and Vibrato

Many singers are not aware of the essential differences between a vibrato and a tremolo. Tremolo is a rapid succession of departure from the true note and returning to it again, so that it becomes difficult to determine what note the singer is trying to sound. A quivering tongue and jaw, as well as a shaky larynx, normally accompany this fault. Bad tremolo may ruin a voice. Possible causes of a tremolo are:

- (a) Using an incorrect breathing method, resulting in an uncontrolled diaphragm.
- (b) Forcing the high notes.
- (c) Undue extension of the registers.

- (d) Overstraining of the voice, either by practising for too long periods at a time, or by trying to force the power of the voice.
- (e) Nervous tension.
- (f) Unsteady tongue and jaw muscles, pulling the larynx against the hyoid bone so that it cannot function freely.
- (g) General fatigue.
- (h) Old age.

Following are some suggestions to cure the problem:

- (a) Refer to breathing exercises, part two, pp. 57-58, and practise the prescribed exercises diligently.
- (b) Refer to register disorders, part two, pp. 90-93, for advice on proper placement of registers.
- (c) Singers should not practise for more than thirty consecutive minutes without time allowed for resting the voice. If directions from the instructor are clear, and the brain is used as well as the voice, a lot will be achieved in that time.
- (d) Refer to exercises for intrusion of the tongue, part two, p. 59, in order to control the tongue.
- (e) Avoid loud singing, and never force the voice.
- (f) Practise singing sustained tones, sung softly and with perfect evenness.
- (g) Often a good rest may help.

Tremolo must not be confused with vibrato. Vibrato does not entail any departure from the sounding note, and, when used correctly, can be one of the greatest assets of the artistic singer. It is an evidence of quality, and a most important feature of all correct and beautiful

singing. It is a characteristic that is evident to the educated listener as a definite sign of a healthy, trained voice.

The necessary control comes from the diaphragm. Inexperienced singers often find it difficult to control the diaphragm. They could practise by raising both arms, bending the elbows to the side of the body, palms towards the face, and the fingers slightly curved. Tense the wrist and hand, and shake them four times to the beat while vocalizing. This movement shakes the breath instead of varying the pitch of the voice. It should not take very long for the diaphragm to learn the art of vibration unaided by the wrist movement.

11.15 Undeveloped Vocal Range

If problems are experienced at the higher end of the vocal range then the singers should be asked to avoid thinking of the high notes as they are approached, because any sense of fear produces rigidity. Rather the singers should concentrate on keeping the throat well open and on retaining the same poise throughout the scale. High notes then become as easy to sing as low ones. To work against the "fear-principle", ask the singers to bend their knees slightly when singing high notes. Each of the following exercises^{1,2,3} should be sung in an even tone without increase in pressure from beginning to end.

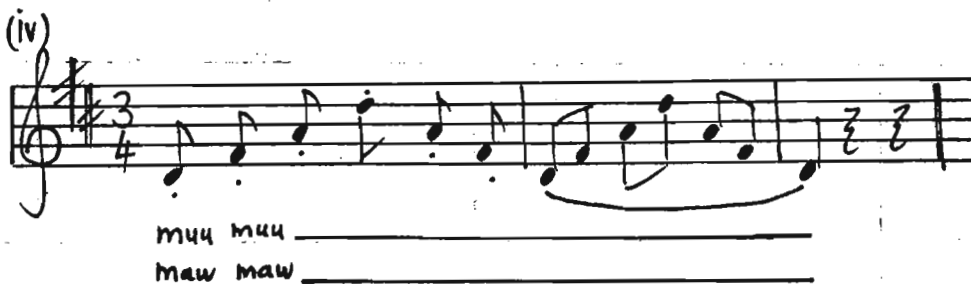
(i)



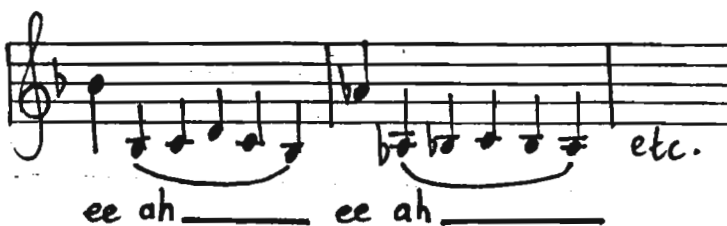
¹Exercise (i): Ivor Warren, *The Examination Candidate's Handbook on Singing*, (London: A. Hammond & Co., n.d.), p. 16.

²Exercise (ii): Judith Litante, *A Natural Approach to Singing*, (London: Oxford University Press, 1962), p. 44.

³Exercises (iii) and (iv): Wilhelm Ehmann and Frauke Haasemann, *Voice Building for Choirs*, (London: Bärenreiter Verlag, 1982), p. 44.



The lower notes of the altos and basses are often weak. The following exercise is designed to improve and strengthen this part of the range. The facial muscles and jaw should be relaxed.



It may happen that during the development of low tones, the singer's high tones do not respond with the usual ease. This is only temporary and should cause little concern. Once the vocal membranes are fully

developed and the throat has become free then the top notes will return with added beauty and brilliance.

11.16 Veiled Tone

A voice is described as being veiled when it sounds muffled, and is not thoroughly clear in quality. This is quite different from a husky tone with which it is often confused. Huskiness is usually due to phlegm collecting on the vocal cords which distorts the sound. Veiled tone is due to a growth on the vocal cords, which produces the muffled tone. The causes are:

- (a) Bad voice production.
- (b) Overworked voice.
- (c) Disease

Certain exercises prescribed for other problems may be of use here:

- (a) Blending of registers, part two, pp. 90-93.
- (b) Staccato exercises, part two, p. 99 (iii) and (iv).
- (c) Breathing exercises, part two, pp. 57-58.

If there is no improvement, the voice should be rested totally and a specialist should be consulted.

11.17 Weak Aural Perception

Many singers are deficient in aural imagination of musical sounds through lack of training. Following are some useful exercises¹ designed to train the choral singer's ear:

- (a) Conduct the choir as it silently sings a familiar song. However, at some predetermined place in the song, the particular note or

¹Wilhelm Ehmann, *Choral Directing*, (Minneapolis: Augsburg Publishing House, 1968), pp. 65-77.

phrase must be sung aloud. The song is then silently resumed, and the process repeated. This promotes concentration and makes the aural imagination alive and alert. Every singer must hear the pitch mentally before singing it. This is one of the most important skills to cultivate for accomplished musicianship. There can be no true sight-reading of music without this skill¹.

- (b) Rehearse without accompaniment whenever possible to improve mental hearing. Continuous use of the piano will hide intonation problems and singers will become aurally dependant on the instrument. A capella singing will:
 - reveal mistakes that the piano covered,
 - create better blend and balance in the group because the singers will be listening to each other instead of adjusting to the piano,
 - force the choir to watch the director more carefully.
- (c) The singing of ascending and descending scale passages is especially beneficial to those singers who are not good at sight reading. The eye will be trained concurrently with the ear as the singer notes the chain of ascending and descending tones and promptly responds to it. Various aids can serve to illustrate this process, e.g. hand gestures, the use of a small ladder or singing by numbers which represent the various degrees of the scale.
- (d) Bass singers in particular must learn to sing descending intervals with confidence and accuracy because these intervals support the whole harmonic structure of a song. A useful method is to sing and identify the first two notes of a well-known song. Descending intervals can be practised by singing the ascending intervals in reverse.
- (e) Melodic dictation trains the ear to grasp the melodic contour and the sequence of notes and helps to strengthen the musical

¹Norman W. Mellalieu, *The Boy's Changing Voice*, (London: Oxford University Press, 1966), p. 31.

memory. It is important to play or sing the tune in its entirety before the students attempt to reproduce it.

- (f) Canon singing trains the ear both melodically and harmonically because the singer listens to the harmony as well as the melody. It also helps the singer to become more self-reliant.
- (g) The building up of chords contributes to better blending and tuning of voices. Each section of the choir is given a note to remember, and after a few moments they are required to sing in harmony together. Another exercise stemming from this one would be to let the singers choose any note of a chord after the root of the chord is given. A more difficult, but extremely beneficial, aural exercise is to sound for example, the pitch A and the singers have to try and find F[#] and D to complete the triad. The pitch A is then given again and the choir is asked to find C and F.

11.18 White Tone

This may be described as a voice without resonance, undirected or unplaced. In singing high notes, male singers often press their chins forcibly on to their necks. This amounts to terrific pressure on the back of the tongue, combined with a sympathetic lifting of the soft palate. These voices have some chest resonance, but little head resonance, and the colour and quality of the voice is "grey", devoid of any ring.

In order to improve the tone, the singer should concentrate on two factors. A foundation of good breath control should be developed and the resonators¹ should be fully used.

11.19 Woolly (Breathy) Tone

The vocal cords are insufficiently approximated which results in an imperfect closure of the glottis. Only a portion of the breath is used to make the vocal cords vibrate, the rest passes through freely. This

¹Refer to the exercises prescribed in part 2, pp. 65-67.

leakage of unvocalized breath, often heard at the commencement of a tone, can be distinctly heard and sounds breathy. Very often this choral tone does not project. Other reasons for this poor tone may be:

- (a) There is inadequate support of the tone.
- (b) The tone is not focused, because the resonators are not fully used.

Following are some suggestions to cure this problem:

- (a) Emphasis should be placed on breath support and control. For example, inhale deeply and exhale on the sound "ff" of the word "piff" for as long as possible.
- (b) Staccato exercises are a good remedy. Sing staccato on "ah" from the tonic to the dominant and back. Gradually ascend and descend in semitones.



- (c) Humming exercises¹ may also help.

¹Refer to resonance exercises in part 2, pp. 65-67.